

6-3c Weighted Average Cost Method

When the weighted average cost method is used in a perpetual inventory system, a weighted average unit cost for each item is computed each time a purchase is made. This unit cost is used to determine the cost of each sale until another purchase is made and a new average is computed. This technique is called a *moving average*.

To illustrate, [Exhibit 5](#) shows the use of weighted average under a perpetual inventory system for [Item 127B](#).

Exhibit 5

Entries and Perpetual Inventory Account (Weighted Average)

Item 127B										
		Purchases			Cost of Goods Sold			Inventory		
Date		Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost
Jan. 1								1,000	20.00	20,000
4					700	20.00	14,000	300	20.00	6,000
10		500	22.40	11,200				800	21.50	17,200
22					360	21.50	7,740	440	21.50	9,460
28					240	21.50	5,160	200	21.50	4,300
30		600	23.30	13,980				800	22.85	18,280
31	Balances						26,900	800	22.85	18,280

The journal entries and the subsidiary inventory ledger for Item 127B are shown in [Exhibit 5](#) as follows:

1. The beginning balance on January 1 is \$20,000 (1,000 units at a unit cost of \$20.00).
2. On January 4, 700 units were sold at a price of \$30 each for sales of \$21,000 (700 units at a selling price of \$30 per unit). The cost of goods sold is \$14,000 (700 units at a unit cost of \$20). After the sale, there remains \$6,000 of inventory (300 units at a unit cost of \$20).
3. On January 10, \$11,200 is purchased (500 units at a unit cost of \$22.40). After the purchase, the weighted average unit cost of \$21.50 is determined by dividing the total cost of the inventory on hand of \$17,200 (\$6,000 + \$11,200) by the total quantity of inventory on hand of 800 (300 + 500) units. Thus, after the purchase, the inventory consists of 800 units at \$21.50 per unit for a total cost of \$17,200.

4. On January 22, 360 units are sold at a price of \$30 each for sales of \$10,800 (360 units at a selling price of \$30 per unit). Using weighted average, the cost of goods sold is \$7,740 (360 units $\times$ \$21.50 per unit). After the sale, there remains \$9,460 of inventory (440 units $\times$ \$21.50 per unit).
5. The January 28 sale and January 30 purchase are recorded in a similar manner.
6. The cost of goods sold for January of \$26,900 is determined by adding the cost of the items sold in the Cost of Goods Sold column.
7. The ending balance on January 31 is \$18,280 (800 units $\times$ \$22.85 per unit).

Check up Corner 6-2

Perpetual Inventory: Weighted Average Method

The beginning inventory, purchases, and sales of Item QX3 for the month of January are as follows:

Jan. 1	Inventory	40 units at \$5
9	Sale	30 units
18	Purchase	70 units at \$7
22	Sale	36 units

The company uses the perpetual inventory system. Determine (1) the cost of goods sold for January and (2) the January 31 inventory balance using the weighted average cost method.

Using Data Analytics

Inventories

A retailer can use data analytics to manage its inventory efficiently and, in doing so, increase its customer satisfaction and profitability. For example, **AutoZone, Inc. (AZO)** operates over 5,000 stores throughout the United States. Each store carries products for a variety of cars, sports utility vehicles, vans, and light trucks. However, stores cannot carry “all” parts for “all” vehicles. Therefore, each store manager must continually update and manage its inventory to best meet the demands (needs) of its local customers.

Data analytics can be used by companies like AutoZone to track inventory and anticipate the needs of their customers. Doing so reduces lost sales due to out-of-stock items and enhances customer satisfaction and a company's reputation for meeting customer needs.

See [TIF 6-5](#) for a homework assignment using data analytics.

Business Insight

Computerized Perpetual Inventory Systems

Your purchases are scanned when you go through the checkout line at **Best Buy Co., Inc. (BBY)**. The scanned data are used to identify the price and adjust the inventory levels. Computerized perpetual inventory systems are used like this when there are many inventory transactions and a manual system is simply not feasible.

Computerized perpetual inventory systems are useful to managers in controlling and managing inventory. For example, if Best Buy has fast-selling items, they can be reordered before the stock runs out. Sales patterns can also be analyzed to determine when to mark down merchandise or when to restock seasonal merchandise. Finally, computerized inventory data can be used to evaluate the effectiveness of advertising campaigns and promotions.